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Behavioral responses to wildfire smoke: Insights from smartphone location data

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Although wildfire smoke is a major public health concern, relatively little is known about individual avoidance behavior. In this paper, we use smartphone-location data to study behavioral responses to smoke during the 2018 California wildfire season, one of the deadliest in recent US history. Using individual-level data, we construct daily measures of when and where California residents spend their time. We find that high levels of smoke increase time spent at home and indoors, while decreasing mobility and time at work. However, we find adaptation starts at relatively high levels of smoke—significantly higher than when smoke becomes dangerous—suggesting a gap between the levels of smoke that trigger adaptation and those that are harmful to human health. While we observe broad-based responses across most demographics, we find much stronger responses among high-SES residents, driven primarily by differences in education rather than income or other characteristics. We compare responses to smoke to those of rainfall, another disamenity to time outside whose costs are more obvious and immediate. While time at home and indoors increases with rain, these responses do not display the same socioeconomic gradient as smoke. Taken together, our results suggest that differential responses by education are unlikely to reflect differences in ability to adjust time and activities. Instead, they likely arise from differences in awareness of the health risks associated with wildfire smoke exposure.

wildfire smoke | climate adaptation | smartphone data | air pollution | behavioral responses

Significance

Estimates suggest that wildfire smoke causes 10 to 16 thousand US deaths annually and this number is growing. Adapting to environmental threats requires behavior change, but studies have long found that responses to related threats differ by education. We find that the college-educated move indoors much more in response to wildfire smoke—but not to rain—than the noncollege educated. This suggests wildfire smoke displays a differential impact by education, which does not reflect a differential ability to move indoors. We also find a significant gap between the level of smoke at which people move indoors and when smoke becomes toxic. These patterns suggest that information and awareness interventions may be key to addressing growing environmental risks.

Wildfire frequency and intensity have been steadily increasing across the globe. In addition to causing billions of dollars in material damage, wildfires generate smoke plumes that can travel thousands of miles while distributing pollutants dangerous to human health, chief among them small particulate matter ($PM_{2.5}$) (1). Wildfire-induced $PM_{2.5}$ pollution has dramatically increased in the United States, threatening to undo years of air quality gains (2). In the past decade, this increase in wildfire smoke is estimated to have caused an additional 10 to 16,000 deaths annually in the United States and 12,000 in California in 2018 alone (3–5). This number is projected to climb to 28,000 by 2050 (4). Monetizing these health impacts, the health costs from wildfire-induced air pollution are projected to be on par with all other US economic costs of climate change, combined (4, 6).

Based on this, mitigating exposure to smoke is crucial, and protection currently depends mostly on private behavioral adaptation. Aside from prescribed burns, policies targeting wildfires focus mostly on air-quality alerts encouraging people to stay indoors. Despite its centrality as a public-health response, evidence on when and how people respond is limited, largely because these kinds of behavioral change are difficult to directly observe.

To address this gap, we create an individual time-use panel dataset based on geolocated smartphone pings for 163,000 devices. Thanks to the high spatial and temporal resolution of these data, we can measure daily time spent at home, at work, indoors, outdoors, and traveling. Our analysis focuses on the 2018 California fire season—one of deadliest and costliest fire seasons on record. We study residents of the Bay Area and Sacramento, a heavily populated geographic zone that was exposed to some of the world's worst air pollution following the 2018 Camp Fire, but which did not receive evacuation orders (see *SI Appendix*, Fig. S1 for area map).

We estimate mobility and time use dose-response functions to wildfire smoke and find strong evidence of heterogeneous avoidance behavior. Our identification strategy rests on the plausibly exogenous duration and path of smoke plumes. Our data allow estimation of dose-response functions across a broad range of smoke concentrations, including

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extreme smoke levels, which are rarely separately studied due to data constraints (7–9). We find that time inside and time at home increase during wildfire smoke events, and that responses are convex—much larger at extreme levels of smoke.

Conversely, time at work and mobility decrease with wildfire smoke. The impact on time at work, at a time when remote work was less common, suggests that labor supply disruptions may explain prior findings of a negative impact of smoke on earnings (1). In addition, we find that individual mobility decreases at moderate to high levels of wildfire smoke, whereas at extreme levels, we observe an increase in long-distance travel away from home, consistent with ref. 9.

We find little responses in time at home and inside to levels of wildfire $PM_{2.5}$ below 50 $\mu g/m^3$, even though these low levels have severe effects on human health, increasing hospital visits for respiratory symptoms (3, 7). This gap suggests that adaptation only starts once smoke becomes salient—for instance through air pollution alerts or visual haze—rather than at the lower concentrations when smoke becomes dangerous. To investigate the relationship between smoke salience and adaptation, we turn to twitter data for a proxy of public awareness, identifying "tweets" related to wildfire smoke and poor air quality. The relationship between smoke-related tweets and smoke $PM_{2.5}$ levels is also strongly convex: Wildfire smoke tweets remain low at mild to moderate concentrations and increase sharply at high smoke levels. These patterns highlight awareness as a key driver of adaptation, and provide evidence on the threshold at which smoke becomes salient to the public.

While many studies have found that underprivileged groups are disproportionally exposed to air pollution such as industrial or automotive emissions, smoke plumes move over large geographic areas in ways that are hard to predict, and which expose demographically and socioeconomically diverse populations (10–14). This feature of wildfire smoke makes it possible to separately study impacts arising from differential exposure and differences in adaptation. Adding to evidence on climate change adaptation disparities (15, 16), we find a strong positive and convex relationship between education and behavioral responses to wildfire smoke.

There may be several reasons for these response gradients. While at high levels wildfire smoke is highly salient, awareness of the health risks associated with smoke exposure may vary. Alternatively, differential ability to adapt could drive heterogeneous responses—lower-income individuals may face more acute time allocation constraints such as less ability to work remotely or scheduling flexibility (17). Put another way, does a strong education gradient in responsiveness to smoke reflect differential awareness of smoke and its dangers, or differential ability to respond?

Consistent with a strong role for awareness, we find that heterogeneity in responses to wildfire smoke is driven almost solely by differences in education, rather than income or other observable characteristics. To further isolate awareness versus ability channels, we also examine dose responses to rainfall, a different disamenity to time outdoors that is widely forecasted, easily observable, and whose costs—such as getting wet—are more transparent. We observe that like smoke, rain increases time inside and at home while decreasing mobility and time at work. In contrast to smoke however, responses to rainfall are largely flat across socioeconomic gradients. If differential ability (say, scheduling flexibility) was the main factor driving differential time reallocation, we would expect similar education gradients for both rainfall and smoke. Taken together, the strong and singular role of education in smoke—but not rain—responses suggests that information and awareness, rather than ability, explain differential responses to smoke.

Results

We find evidence of avoidance behavior: When exposed to wildfire smoke, individuals reduce their mobility and spend more time at home, inside, and less time at work. Responses to moderate but dangerous smoke $PM_{2.5}$ are much smaller than to higher, more salient levels (Fig. 1 and Table 1). Compared to low-pollution days (smoke $PM_{2.5}$ between 0 and 5 $\mu g/m^3$), on extreme days (above 150 $\mu g/m^3$) time at home increases by 40 min, or 2.4 percentage points. Similarly, on extreme smoke days, time spent inside increases by 26 min, a 1.1 percentage point increase, and time spent at work decreases by 19 min, a 1.8 percentage point decrease (Table 1 and *SI Appendix*, Table S3). Daily mobility declines by 3 to 5% on days with moderate to high smoke. We find no impact of extreme smoke on daily mobility, but detect a 0.3 percentage point increase in trips further than 100 km from home in weeks with very high concentrations (*SI Appendix*, Table S5), suggesting that at high levels of smoke some avoidance in-place behavior transitions to temporary relocation (9). Our estimates are comparable in magnitude to other studies of air pollution in different contexts: ref. 18 find that on days with (nonwildfire) $PM_{2.5} > 75$ $\mu g/m^3$, hours worked are reduced by 30 min. Closest to our work, on high smoke days, ref. 8 find a 3 percentage point increase in the percentage of people entirely at home and ref. 7 observe a 3 percentage point drop in recreation POI visits.

Because wildfire smoke can carry health risks at levels below where it becomes conspicuous, we investigate and do not find evidence of strong avoidance behavior for low to moderate levels of wildfire smoke. Our binned specification results suggest that individuals start to increase time at home and inside once smoke crosses 50 $\mu g/m^3$, while the magnitude of the response to smoke for time at work and distance traveled also increases around this threshold (Table 1). The spline specification shows similar patterns for time at home and inside, with responses only starting to pick up above 35 $\mu g/m^3$. However, ref. 7 find that even moderate levels of wildfire smoke have adverse health effects, with smoke levels between 10 and 25 $\mu g/m^3$ causing a 10% increase in emergency department (ED) visits for acute respiratory conditions, and even a 25% increase in asthma-related visits in California. There is therefore a measurable discrepancy between the level at which smoke becomes dangerous to human health and when people start to avoid exposure. Our finding of limited avoidance behavior at low levels of smoke reinforces other evidence suggesting that recreation visits and traffic drop off only for smoke levels above 50 and 25 $\mu g/m^3$ respectively (7).

We compute dose responses to rainfall both to better understand the smoke-response mechanism, and to test our ability to detect time-use responses with a different disamenity to time outside. On days with moderate to heavy rainfall (above 30 mm) time at home and inside increase (respectively by 1.8 and 1.1 percentage point) while time at work and mobility decrease (by 1.5 percentage points and 19%, see Table 1). These time-use patterns are consistent with expected responses to rainfall, and of a similar magnitude to wildfire smoke responses.

We explore whether responses differ across socioeconomic groups and find a strong education gradient in responses to wildfire smoke, with education appearing more important

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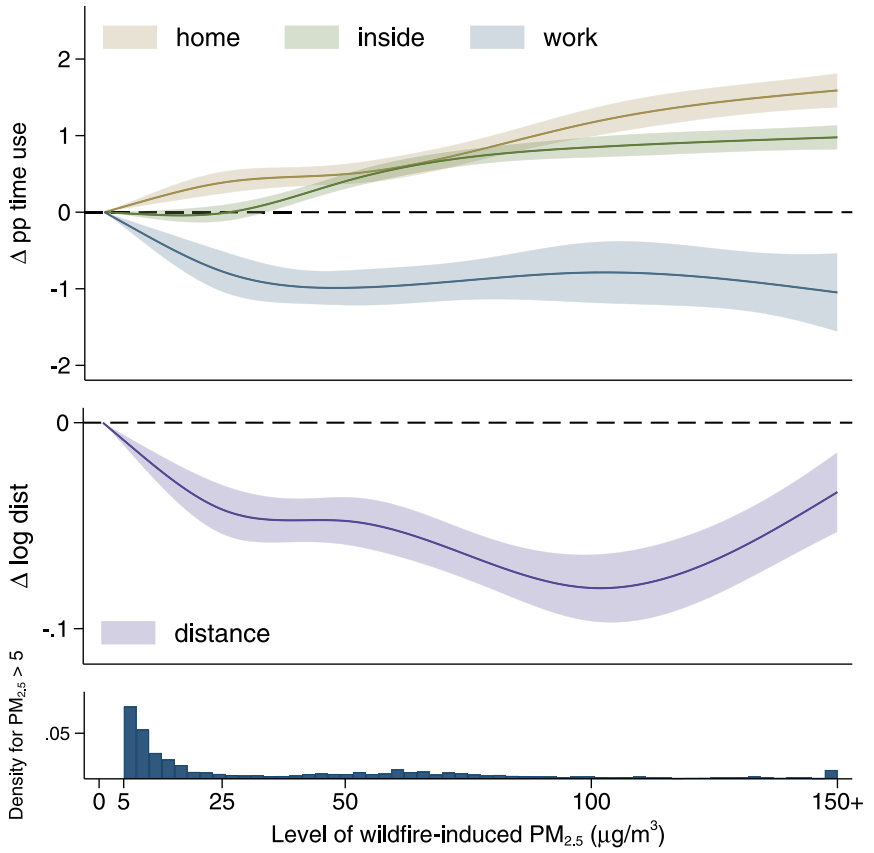

Fig. 1. Time use dose responses to wildfire smoke–spline specification. The graph shows predicted values from a linear regression with cubic splines using knots at 5, 25, 50, 100, and 150 μg/m^3. For each value of smoke $PM_{2.5}$ between 0 to 150 μg/m^3, we compute the predicted effect by evaluating the spline basis and taking a linear combination with the estimated coefficients. Controls include quadratics for temperature and precipitation, day-of-week, device, and month fixed effects. SEs are derived using the delta method. The density plot omits observations with smoke between 0 and 5 μg/m^3.

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than other socioeconomic characteristics, including income, in shaping responses. While highly correlated (*SI Appendix*, Fig. S6), disentangling the impacts of education and income may be crucial for smoke exposure mitigation policies, as income may map more to differences in ability to respond, while a direct effect of education is more consistent with information and awareness and channels.

We construct 75 overlapping 25% percentile sample windows, based on the distribution of the share of college-educated and median household income. We infer a device's socioeconomic status by matching its home location to census block-group data. We re-estimate our baseline equation on each 25% window, starting with the bottom quartile and moving incrementally to the top of the distribution. We find considerable socioeconomic gradients in responses to high levels of wildfire smoke (above 150 μg/m^3, see Fig. 2 and *SI Appendix*, Fig. S8). When exposed to extreme smoke, the quartile with the highest share of college-educated residents increases their time inside by 2.5 percentage points more than the bottom quartile. We find similar results for income, although the difference is more muted. For time at home, the education gradient is even stronger, with responses in the top quartile 5 times larger than in the bottom quartile, while we find little evidence of an income gradient. Consistent with the findings of ref. 9, we also find that the richest and most educated are more likely to temporarily relocate during wildfire smoke events (*SI Appendix*, Table S5).

To examine the relative importance of income and education, we test for demographic heterogeneity and consistently find that the impact of education dominates all other observable covariates (Table 2 and *SI Appendix*, Table S10). We also consider the joint effect of education and income by extending the quantile analysis in Fig. 2 panels (*v* and *vi*). Within income quantiles, high education appears to increase responses to wildfire smoke, in particular in low-income areas. In contrast, high income appears to have a null or even slightly negative effect within education quantiles.

Education also matters more than other demographic characteristics such as race, ethnicity, age, and urban–rural classification. Individuals in urban areas tend to adapt more, while individuals in areas with a high share of Hispanic or young residents tend to respond less, consistent with existing evidence on environmental inequalities (*SI Appendix*, Table S10) (10, 11, 13, 14). Education has a large effect among all heterogeneity dimensions considered. Next, we jointly estimate the impact of education and other dimensions pairwise, and find a positive and consistent effect of education across specifications (1.3 to 1.4) percentage points), with the independent impact of other dimensions becoming indistinguishable from zero, including for income (Table 2). Finally, the only covariate selected by a LASSO is the share of college graduates, further confirming our finding (*SI Appendix*, Table S9).

To decompose the role of ability and awareness, we also examine the relationship between adaptation and socioeconomic status to rainfall. The limited potential for differential awareness of rainfall allows differences between smoke and rain to disentangle ability and information channels. We find much stronger educational heterogeneity in responses to smoke than to rainfall, which displays almost no education gradient (Fig. 2

Table 1. Dose-responses to smoke and precipitation

Panel A: Dose responses to Smoke

	(1) %Time at home	(2) %Time inside	(3) %Time at work	(4) Log(Distance)
5 to 25 μg/m^3	0.221***	−0.058	−0.567***	−0.029***
	(0.058)	(0.044)	(0.099)	(0.004)
25 to 50 μg/m^3	0.055	0.070	−0.550***	−0.006
	(0.109)	(0.083)	(0.191)	(0.008)
50 to 100 μg/m^3	0.889***	0.713***	−0.940***	−0.052***
	(0.089)	(0.068)	(0.142)	(0.006)
100 to 150 μg/m^3	0.934***	0.837***	−0.617**	−0.041***
	(0.143)	(0.106)	(0.284)	(0.011)
150+ μg/m^3	2.378***	1.123***	−1.882***	−0.022
	(0.221)	(0.157)	(0.425)	(0.017)
N	4,771,539	4,771,539	876,078	4,771,539
R^2	0.441	0.365	0.446	0.517

Panel B: Dose responses to precipitation

	(1) %Time at home	(2) %Time inside	(3) %Time at work	(4) Log(Distance)
10 to 30 mm precip.	0.771***	0.531***	−2.051***	−0.118***
	(0.047)	(0.035)	(0.087)	(0.003)
Above 30 mm precip.	1.793***	1.094***	−1.533***	−0.190***
	(0.121)	(0.091)	(0.244)	(0.009)
Temp. and prec. controls	yes	yes	yes	yes
Month, device FE	yes	yes	yes	yes
Day-of-week	yes	yes	yes	yes
# of devices	162,558	162,558	23,677	162,558
N	4,771,539	4,771,539	876,078	4,771,539
R^2	0.441	0.365	0.446	0.517

Notes: *$P < 0.10$, **$P < 0.05$, ***$P < 0.01$. SE in parentheses and clustered at the device level. Device-day observations that travel less than 300 km during the day, whose home is not within 50 km of an active wildfire, and for which we observe at least 10 pings, 15 h of the day, and 10 h with at least one ping are included. The baseline category is pollution between 0 and 5 μg/m^3 for smoke, and days with rainfall between 0 and 10 mm for responses to precipitation. For time at work, we restrict to weekdays and to devices for which we have an identified workplace.

and *SI Appendix*, Table S6 and Fig. S8). That responses to rain show no education gradient suggests that the college educated are not strongly differentially able to shift time indoors. This makes differential awareness of the dangers of smoke a much more plausible explanation of the strong education gradient in smoke response.

Discussion

Our finding of a strong and singular effect of education helps integrate several important existing findings about wildfire smoke. A large literature finds that information provision encourages the adoption of protective air-pollution behavior (19–21). However, previous papers have found information availability does not show many of the demographic gradients present in behavior or harms; for example internet searches about wildfire smoke do not vary across the income gradient (8).

Our results help reconcile these findings. Wildfire smoke appears to typify a set of health behaviors described by what health economists call the allocative efficiency hypothesis—that education directly improves the ability to process and act on health-related information (22). This literature has documented strong education gradients in several health decisions, most notably cancer screening and cigarette smoking. For example between the 1970s and 2000s, as both US female college attendance and evidence regarding the dangers of tobacco smoke grew, smoking increased for noncollege educated females, but dropped for both male and female college attendees (12). Similarly, college-educated women both receive more mammograms, and increase their mammogram rates more in response to the cancer diagnoses of blood relatives (22). Prior work shows that differences in receptivity to scientific evidence are at least partially responsible for these education gradients—our work suggests this may also drive disparate smoke avoidance, especially since beliefs in the severity of climate change differ by educational attainment (23, 24).

The avoidance behavior patterns we estimate help explain findings of concave, even nonmonotonic health impacts of wildfire smoke. For example, refs. 3 and 7 find that ED visits rise sharply at low levels of smoke, but then quickly level off and even decline at high smoke levels. This is consistent with the strongly convex avoidance behavior patterns we find if protective measures can effectively prevent ED visits.

Our finding of a sharp increase in behavioral responses around 50 μg/m^3 suggests that awareness of smoke likely jumps at around this level. To assess whether this jump in the salience of smoke can be independently observed, we study the universe of "tweets" posted from Northern CA in our sample period, and ask how mentions of smoke increase with smoke. Previous work has shown that twitter data can be used for sentiment analysis (8, 25) and to detect wildfire smoke events (26). Our approach asks when smoke becomes conspicuous and salient enough to prompt sharing.

We use a machine-learning classifier to identify tweets mentioning air pollution or wildfire smoke, and find that the share of smoke-related tweets is low and flat below 35 μg/m^3, then

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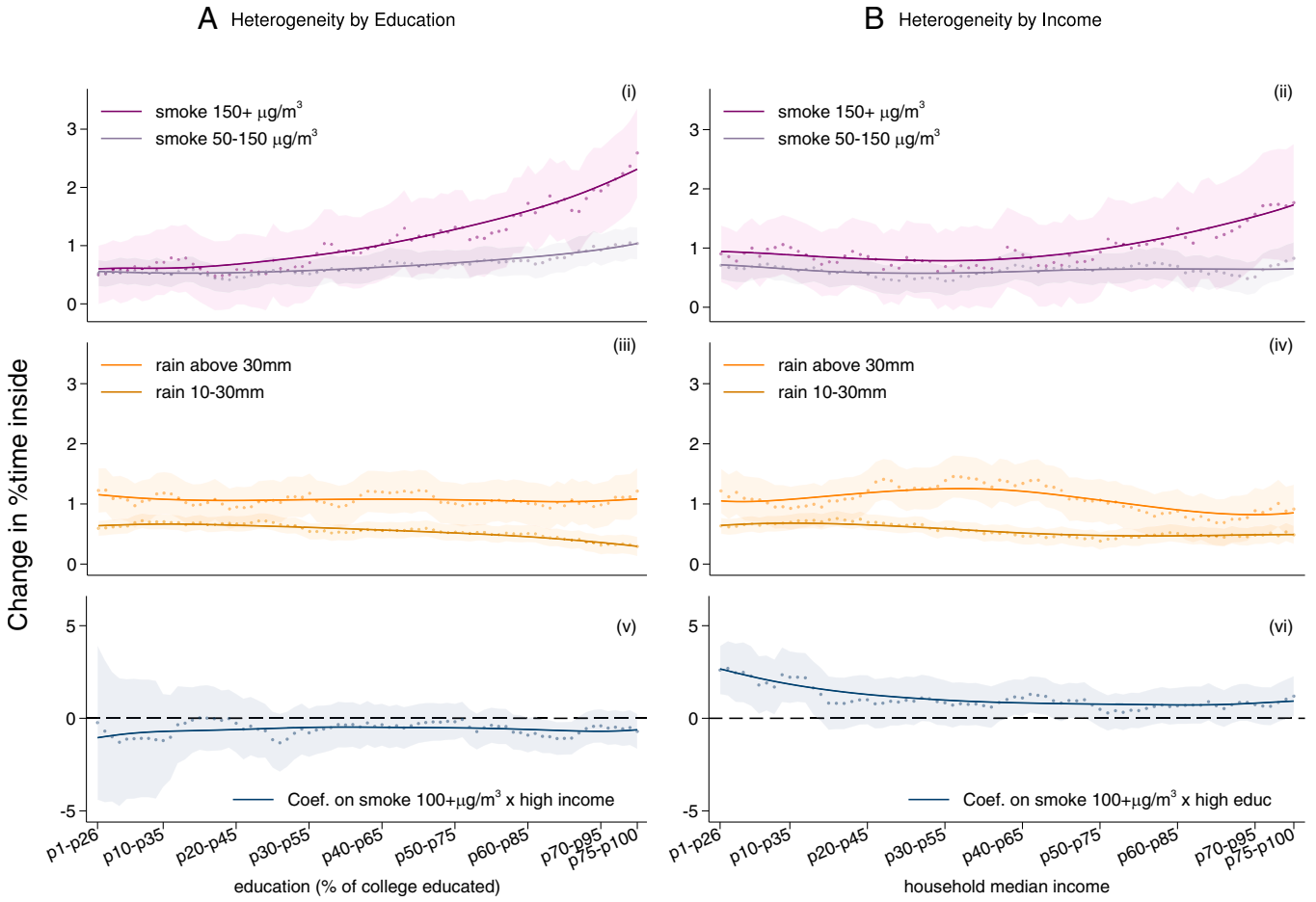

Fig. 2. Responses across socioeconomic groups for time inside. Each plot represents plots coefficients for 75 regressions. In panels (*i–iv*), each point represents a regression coefficient from estimating Eq. **1** with 3 smoke or precipitation bins and restricted to a sample according to education or income level from the block-group level ACS. For income we use median household income and for education the percentage of college-educated individuals. We split the sample in 75 quantiles of the income and education distributions, starting with the bottom 25% and incrementally moving to the top 25%. Panels (*v* and *vi*) adopt the same quantile-based approach but estimate Eq. **2** with $k = 1$. For each quartile of the education distribution, we measure the impact of high income (*v*), and the impact of high education for each quartile of the income distribution (*vi*). The dependent variable is time spent inside. We fit a nonparametric locally smoothed curve between estimated coefficients and quartile rank.

rises sharply between 40 and 50 μg/m^3, continuing to rise well past 125 μg/m^3 (Fig. 3). The frequency of smoke-related tweets increases by a factor of 38 between low (5 to 10 μg/m^3) and extreme (150+ μg/m^3) smoke days. This strong convexity mirrors our measured behavioral responses, reinforcing that the lack of smoke salience—likely before it is sensed or seen—may explain the lack of adaptation below 35 μg/m^3. Existing work also suggests that between 35 and 50 μg/m^3 mild respiratory symptoms begin to occur across population groups, presumably making the presence of smoke hard to ignore (27). Corresponding thresholds in warning systems may also contribute; the EPA 24-h nonattainment standard is 35 μg/m^3, and $PM_{2.5}$ pollution over 35 μg/m^3 corresponds to an AQI over 100 ("unhealthy for sensitive groups") while $PM_{2.5}$ over 55 μg/m^3 corresponds to an AQI over 150 ("unhealthy"); these thresholds trigger government warnings and alerts (27, 28). The observed increases in tweets at these levels are consistent with prior evidence that these alerts are broadly seen and salient (21, 29).

Our work also helps explain findings on the economic impacts of wildfire smoke exposure (1). Disaggregating the quarter-county level economic impact of smoke days, we find that extreme wildfire smoke decreases daily hours worked (observed at the office) by roughly 4 to 5%. This decrease in hours worked likely reflects both mobility-reducing adaptation and smoke-related illnesses and absenteeism, which are the mechanism most noted in the literature on economic impacts of air pollution (30, 31). Since we measure physical office presence, we cannot observe the degree to which smoke induces a switch to remote work, though our analysis predates COVID and the widespread adoption of remote office work. We do find that devices residing in richer areas decrease office hours more than average (*SI Appendix*, Fig. S8). This could reflect differences in smoke health-risk awareness or air filtration at work, in the feasibility of remote work, in scheduling flexibility, or other differences in the opportunity cost of staying home (17, 32).

One limitation of our focus on time use is that staying indoors may offer only partial protection; studies of air-quality monitor data find high smoke infiltration rates (8, 33). Other behaviors requiring investments, such as air filtration, may require both an awareness of the dangers of smoke and financial resources. Consistent with this possibility, we find that unlike most time-use metrics we study, income remains a significant predictor of temporary relocation during smoke spells (*SI Appendix*, Table S5). However, our findings suggest a central role for awareness in both the muted response to low levels of smoke, and the

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Table 2. Heterogeneity: Pairwise comparisons of education versus other dimensions

Dep var: %time inside	(1) Educ.	(2) Income	(3) Urban	(4) Black	(5) White	(6) Asian	(7) Hispanic	(8) Age > 75	(9) Age < 18
Smoke > 150 μg/m^3	0.605*** (0.172)	0.598*** (0.175)	−1.910 (1.554)	0.702*** (0.251)	0.578*** (0.187)	0.564*** (0.181)	0.725*** (0.202)	0.656*** (0.191)	0.813*** (0.202)
Smoke > 150 μg/m^3 × het. var. > 75th pct.	1.390*** (0.449)	0.130 (0.565)	2.557 (1.565)	−0.179 (0.329)	0.142 (0.387)	0.350 (0.480)	−0.495 (0.384)	−0.246 (0.379)	−0.592* (0.356)
Smoke > 150 μg/m^3 × educ > 75th pct.		1.357*** (0.461)	1.347*** (0.449)	1.331*** (0.463)	1.336*** (0.475)	1.378*** (0.449)	1.270*** (0.462)	1.402*** (0.449)	1.254*** (0.451)
Temp. and prec. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE, device FE, day-of-week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	yes
# of devices	155,719	155,719	155,719	155,719	155,719	155,719	155,719	155,719	155,719
N	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730
R^2	0.364	0.364	0.364	0.364	0.364	0.364	0.364	0.364	0.364

*Notes: *$P < 0.10$, **$P < 0.05$, ***$P < 0.01$. SE in parentheses and clustered at the device level. Smoke > 150 μg/m^3 is an indicator for extreme smoke pollution days. "het. var. >75th pct." is an indicator equal to 1 if the device is within the top quartile of the specific heterogeneity dimension listed at the top of each column, where each characteristic is assigned based on the Census block group of the device home location. Device-day observations that travel less than 300 km during the day, whose home is not within 50 km of an active wildfire, and for which we observe at least 10 pings, 15 h of the day, and 10 h with at least one ping are included. Sample is restricted to devices for which there is median block group household income information in the American Community Survey. The baseline smoke $PM_{2.5}$ category is pollution between 0 and 150 μg/m^3.

strong education gradient in responses at all levels—wildfire smoke appears to be an important example of the allocative efficiency hypothesis. Unless fires are prevented themselves, these awareness gaps likely pose a barrier to all behavioral interventions, and bridging these gaps may be crucial to addressing the growing harm of wildfire smoke.

Materials and Methods

Wildfire-Induced $PM_{2.5}$. We use the ref. 2 dataset to measure daily $PM_{2.5}$ produced by wildfires on a 10 × 10 kilometer grid. This dataset uses machine-learning to predict $PM_{2.5}$ concentrations from different sources, including ground-truth measurements through EPA monitors as well as remotely sensed and reanalysis products.

In our device-day sample, the average wildfire-induced $PM_{2.5}$ exposure is 4.1 μg/m^3 (Table 3). The median and 75th percentiles are 0, reflecting that on most days there is no wildfire-induced pollution. We obtain 157,000 device-day observations above 50 μg/m^3–a large number of observations to study behavioral responses to extreme levels of pollution thanks to the granularity of the smartphone data.

Temperature and Precipitation. Daily rainfall and daily minimum and maximum temperature come from the gridded PRISM weather product with a fixed number of stations at the daily, county level (34).

Socioeconomic and Demographic Data. We use publicly available socioeconomic information from the 2020 US Census and American Community Survey 5-y averages (2016–2020) at the Census Block Group level, which corresponds to 1,500 residents on average. We also use the USDA rural–urban classification at the tract level and classify tracts as "urban" if they are categorized as "metropolitan area core" or "metropolitan area high commuting" (35). We assign socioeconomic characteristics to each device based on the Census Block Group of its inferred home location.

Smartphone Mobility Data. We use individual smartphone ping data collected by Veraset to construct time-use metrics. These data come from recorded device geocoordinates by phone operating systems and consist in an anonymized set of pings by device, complete with geocoordinates, timestamps, and GPS margins of error (usually a few meters).

In this dataset, we have 163K devices from March 2018 to February 2019 for which we can construct daily time use metrics. We obtain an unbalanced panel of individual, daily time use. We find that this sample is overall spatially representative, covering on average and at the median 1.6% of each block group in the area of interest (*SI Appendix*, Fig. S1). There is a small negative relationship between median household income, number of college graduates, and number of devices identified in a given block group (*SI Appendix*, Table S1). Previous work using this dataset has also found it to be relatively representative (36–39).

Methodology for Time-Use Metrics. We leverage the richness of the cell phone ping dataset to construct daily time allocation at the individual level. We do not observe any characteristic at the individual level other than cell-phone pings.

Home location. the modal location where the device pings at night during the third quarter of 2018.

Work location. the modal location where the device pings during weekdays during business hours in November 2018. Since the methodology to identify workplace is based on repeated visits to a building during business hours, we can only identify office workers with a regular schedule, leaving out gig and outdoor workers. Devices with a determined office location reside therefore on average in wealthier areas than the rest of the sample (*SI Appendix* Table, S1).

Outdoor versus indoor. We overlay ping geolocation with a raster of buildings (computed before our wildfire period) to determine whether each observed ping falls outside or inside of a building. This last measure is more subject to error as it involves both GPS error and inference from satellite maps, but these errors are likely to be classical and time-use is an outcome variable. Extreme wildfire smoke may be correlated with increased GPS error, which might bias this measure toward time-outdoors and attenuate our measured effect.

Distance. We compute daily distance traveled by computing the distance between each ping's location and its previous location, and summing by day and device.

First, we label pings into categories (home, work, outside, inside). Next, we interpolate to transform the discrete list of pings into a continuous measure of time use based on a mix of nonparametric and parametric assumptions (*SI Appendix*, Fig. S3). Importantly, we do not constrain any of the time-use metrics: The number of observed hours on a given day does not need to sum up to 24. On a given day, the number of hours observed can vary by device (*SI Appendix*, Fig. S3).

Data quality. Applying this methodology, we obtain estimates of time allocation at the device-day level. We find no significant variation in hourly coverage–measured as the number of hours with at least one observed ping–and in

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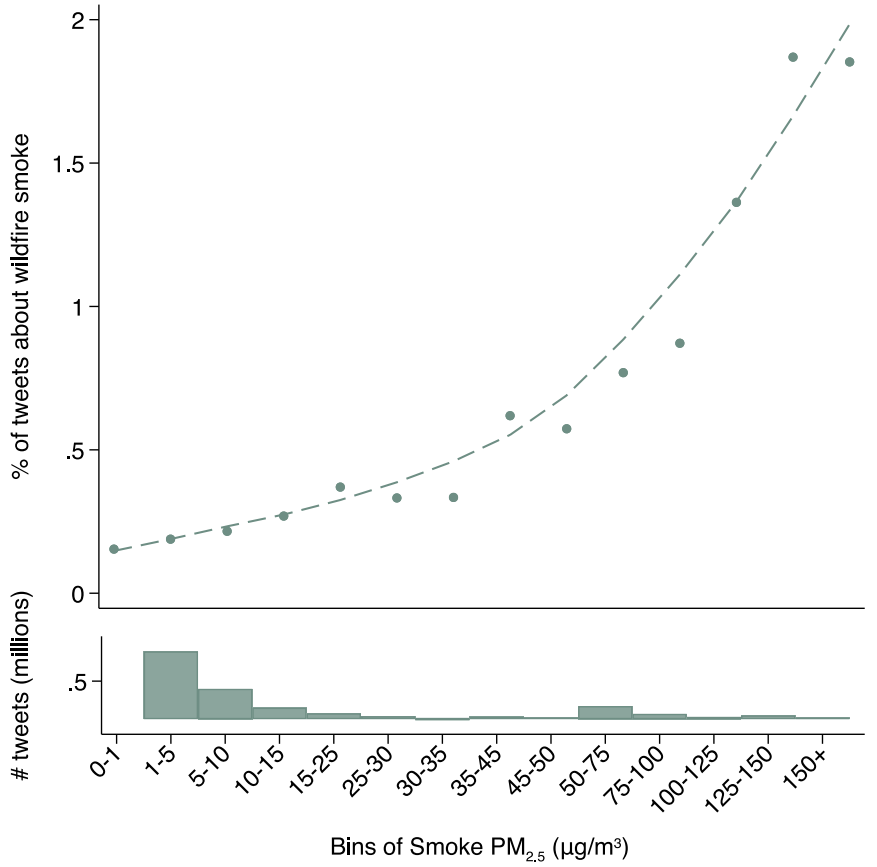

Fig. 3. Salience of wildfire smoke in Twitter data for different smoke levels, detailed binning Note: The *Top* panel shows the distribution of tweets across $PM_{2.5}$ smoke concentration bins. The main graph shows the percentage of wildfire smoke-related tweets by smoke level, with a nonparametric trend fit. Tweets about wildfire smoke are identified using a Logistic Regression classifier trained on manually labeled tweet data and rule-based labels. Text features are extracted using TF-IDF vectorization. The *Bottom* panel shows the distribution of smoke is shown for smoke > 0 since most days have no smoke.

constructed time use hours across smoke levels (*SI Appendix*, Table S11). In the final time use sample, we restrict to device-day observations with sufficient ping density to construct daily time use, defined as at least 10 pings per day, coverage in at least 10 h, and at least 15 h of constructed time use (which can exceed raw coverage due to the buffering process). In addition, we also filter out observations that are within a 50 km radius of an active wildfire to distinguish the impact of wildfire-induced air pollution from the wildfire itself. For each device and day, we match home location to county-level centroid-based distance to nearest fire from ref. 8. Last, we restrict the time-use analysis to devices that travel less than 300 km in a day.

Comparison with ATUS. We find that for devices meeting those conditions, the average daily time spent inside is 18.2 h, time spent at home 14.8 h, and time spent at work 7.1 h (*SI Appendix*, Table S2). We compare our time use metrics to the reference dataset for time use in the US, the American Time Use Survey [restricting the ATUS sample to California from 2010 to 2018 for comparability, (40)]. The ATUS is a cross-sectional survey where respondents report a time-use diary of their daily activities; the survey does not specifically ask about time spent indoors or at home versus elsewhere. We follow the methodology of ref. 41 to construct time spent outdoors from the ATUS, and adopt a similar approach for activities done at home. We find that our time-use estimates are mostly consistent with the ATUS (*SI Appendix*, Table S2). Time spent outside is longer in the smartphone data than in ATUS, likely due to both measurement challenges in ATUS and potential overestimation in our dataset–perhaps because of GPS error, but this should not affect our estimates as long as errors are random.

Estimating Responses to Wildfire Smoke. To estimate causal dose responses to smoke $PM_{2.5}$, our underlying assumption is that air pollution generated by wildfire smoke constitutes an exogenous shock because the timing, intensity, and direction of plumes is hard to predict. In addition, smoke plumes cover extensive areas: The $PM_{2.5}$ concentrations they generate are unlikely to vary within commuting zones or cities, creating blanket pollution shocks. We therefore assign individual, daily smoke $PM_{2.5}$ based on device home location.

Our main specification is the following (Table 1 panel A):

$$y_{it} = \alpha + \sum_k \beta_k binPM_{skt} + \mathbf{X_{it}} + \omega_i + \omega_m + \omega_{dow} + \epsilon_{it} \quad [1]$$

y_{it} is a time use metric or log distance traveled and varies at the daily (t) and device level (i). We split smoke $PM_{2.5}$ into 6 bins ($binPM_{skt}$): 0 to 5, 5 to 25, 25 to 50, 50 to 100, 100 to 150, and 150+ $\mu g/m^3$, with the 0 to 5 $\mu g/m^3$ bin as the reference bin. $PM_{2.5}$ device exposure level is assigned based on the tract level s of device home location. We include quadratics in daily average temperature and precipitation as controls. Device fixed effects ω_i, month fixed effects ω_m, and day-of-week fixed effects w_{dow} are included. SEs are clustered at the device level, but results are robust to clustering at day × tract, see *SI Appendix*, Table S3. We use a similar specification for rainfall and control for quadratics in temperature and smoke $PM_{2.5}$. For rainfall, we consider daily precipitation bins of 0 to 10 mm (light rain), 10 to 30 mm (moderate rain), and above 30 mm (heavy rain). Conditioning on positive values, the 30 mm and 150 $\mu g/m^3$ cutoffs for rain and smoke correspond approximately to the 99th percentile of their respective distributions. In *SI Appendix*, Fig. S8, we also verify that rainfall heterogeneity results are robust to an alternative threshol of 50 mm.

To estimate heterogeneous responses to smoke across socioeconomic groups, we estimate the following:

$$y_{it} = \alpha + \sum_k \beta_k \text{ bin } PM_{skt} + \sum_k \zeta_k \text{ bin } PM_{skt} \times \mathbf{1}_{\text{het. var.}>75\text{th pct}} + \mathbf{X_{it}} + \omega_i + \omega_m + \omega_{dow} + \epsilon_{it} \quad [2]$$

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Table 3. Summary statistics

	Mean	SD	Min/Max
Hours inside	18.2	5.2	0/24
Hours at home	14.8	6.7	0/24
Hours at work (cond. on working)	7.1	4.0	0/24
Daily total hours observed	21.3	2.7	15/24
Daily distance traveled (km)	44.3	55.1	0/300
%Time inside	84.4	19.8	0/100
%Time at home	68.1	27.8	0/100
%Time at work (cond. on working)	34.0	19.1	0/100
Smoke $PM_{2.5}$ ($\mu g/m^3$)	4.1	17.3	0/241
Rainfall (mm)	2.1	6.0	0/81
Distance from active wildfire (km)	181.5	148.6	50/1512
Max distance from home (km)	52.5	335.9	0/9437
Median household income (in 1000s USD)	98.4	48.5	2.5/250.0
% College educated	0.4	0.2	0/1

Smoke level, $\mu g/m^3$	# of obs (in 1000s)	rainfall, mm	# of obs (in 1000s)
0 to 5	4,325	0 to 10 mm	4,456
5 to 25	262	10 to 30 mm	295
25 to 50	50	above 30 mm	42
50 to 100	113		
100 to 150	32	Total # of obs	4,793
150+	12		

Eq. **2** measures the interacted effect of smoke and other socioeconomic or demographic characteristics including education, income, race and ethnicity, age, and urban–rural status. We reduce the number of $PM_{2.5}$ bins for heterogeneity analyses compared to the baseline. In *SI Appendix*, Table S6 and Fig. 2, we keep $k = 3$ bins of $PM_{2.5}$ equal to 0 to 50, 50 to 150, and 150+ $\mu g/m^3$, whereas 0 to 5 $\mu g/m^3$ is the reference bin in our baseline specification. In *SI Appendix*, Tables S7, S8, and S10 and Fig. 2 panels (*v* and *vi*), we set $k = 1$ and vary the bin partition. $\mathbf{1}_{\text{het. var.}>75\text{th pct}}$ is an indicator for being above the 75th percentile in the distribution of a characteristic of interest. We estimate similar specifications for heterogeneous responses to rainfall.

To allow for a flexible relationship between wildfire $PM_{2.5}$ and time use, we also estimate a cubic spline specification with 5 kn placed at 5, 25, 50, 100, and 150 $\mu g/m^3$ (Fig. 1). Each outcome is regressed on the smoke spline terms with the same controls and fixed effects as in Eq. **1**. For each value of smoke $PM_{2.5}$ between 0 to 150 $\mu g/m^3$, we compute the predicted effect by evaluating the spline basis and taking a linear combination with the estimated coefficients. SEs at each level are derived using the delta method.

Travel Away From Home. For each ping, we compute the distance between its geocoordinates and the identified home location of the device and take the daily maximum by device. Since traveling to avoid air pollution could happen before the smoke event, we collapse the dataset to the weekly level (w). The dependent variable $\mathbf{1}_{i,w\text{ far away}}$ is a weekly binary indicator measuring whether the device was observed being at least once more than 50, 100, or 150 km away from home. This outcome is regressed on a binary variable $\mathbf{1}_{w,\text{ smoke }PM_{2.5}>150}$ indicating whether smoke crossed the 150 $\mu g/m^3$ threshold at least once that week in the device home census tract. Device and month fixed effects ω_i, ω_m are included.

$$\mathbf{1}_{i,w\text{ far away}} = \beta \mathbf{1}_{w,\text{ smoke }PM_{2.5}>150} + \omega_i + \omega_m + \epsilon_{iw} \quad [3]$$

Identifying Tweets About Wildfire Smoke. Our analysis uses the geocoded database of tweets constructed by ref. 42 for the year 2018. We apply a supervised machine learning approach to identify tweets about wildfire-induced air pollution. Given that most tweets are approximately geocoded at the city-level, we restrict to tweets within our area of interest (see *SI Appendix*, Fig. S1 for bounding box). This creates a dataset of about 11 million tweets for March through December of 2018. To train the model, we create a labeled set of tweets in two steps. First, we manually label 500 tweets from November 16, one of the smokiest days in our period. Second, we include in the training set tweets that explicitly reference wildfire smoke or poor air quality (i.e. "wildfire smoke," "bad air quality," "aqi") and a list of "false positives," i.e. terms that are linguistically related but refer to other topics ("I smoke," "air pods," "air fryer," etc.). Using TF-IDF vectorization to identify word patterns associated with wildfire smoke tweets, we train a logistic regression classifier on the labeled sample.

Data, Materials, and Software Availability. Replication data has been posted in the Harvard Dataverse (43).

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Supporting Information for

Behavioral Responses to Wildfire Smoke: Insights from Smartphone Location Data

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This PDF file includes:

Figs. S1 to S8
Tables S1 to S11
SI References

Supplementary Information

Supplemental Information Regarding Time-Metrics Methodology . We make some parametric and non-parametric assumptions in order to change discrete pings into continuous time use. To do so we must make assumptions about what happens between two observe pings. First, we consider that a device is traveling if the elapsed time and distance between two pings leads to a speed above 10km/hour. The first assumption is a non-parametric one. Consider that a device is not traveling and is observed at time x at location i and a ping at time $x+1$ at location j. We consider that this elapsed time is split evenly between i and j if the device is not travelling. For instance, on figure S3 example 1, this assumption means that for the one hour elapsed between ping 4 at home and ping 5 at work, we assign 30 minutes to home and 30 minutes to work. We also make a couple of parametric assumptions. For instance, if the first ping of the day is before 9AM in the home, we consider that the period between midnight and the first ping are spent at home. Similarly, if the last ping of the day is after 8PM within the home, we consider that the time remaining between the last ping and midnight is also spent in the home. We also assume that time spent in the office is a single interval. This means that we compute time spent in the office as time elapsed between the first ping of the day in the office and last ping of the day in the day in the office. That is, if the person goes out for lunch during her workday or attends a meeting in a different building, this will still be considered work time.

Robustness Checks. In addition to the analyses presented in the main text, we verify that our baseline results on the impact of wildfire smoke on time use and mobility are robust to alternative specifications, including alternative clustering choice (see Table S3) and alternative $PM_{2.5}$ bin choices (see Table S4).

We also check that our results on the importance of education do not depend on selected smoke $PM_{2.5}$ cutoffs in Table S7. In addition, since income and education are the most correlated characteristics in our sample (see Figure S6), we also check that we have sufficient support to disentangle their respective impacts. To identify spatial variation in the joint distribution of income and education, we rank each block group by education and income and map neighborhoods in the top and bottom quartiles of the difference between these ranks (see Figure S7). Because smoke plumes blanket large geographic areas (see Figure S2), areas exposed to the same level of pollution contain neighborhoods that rank high in the share of college educated residents but low in the income distribution, and vice versa.

Fig. S1. Device Sample

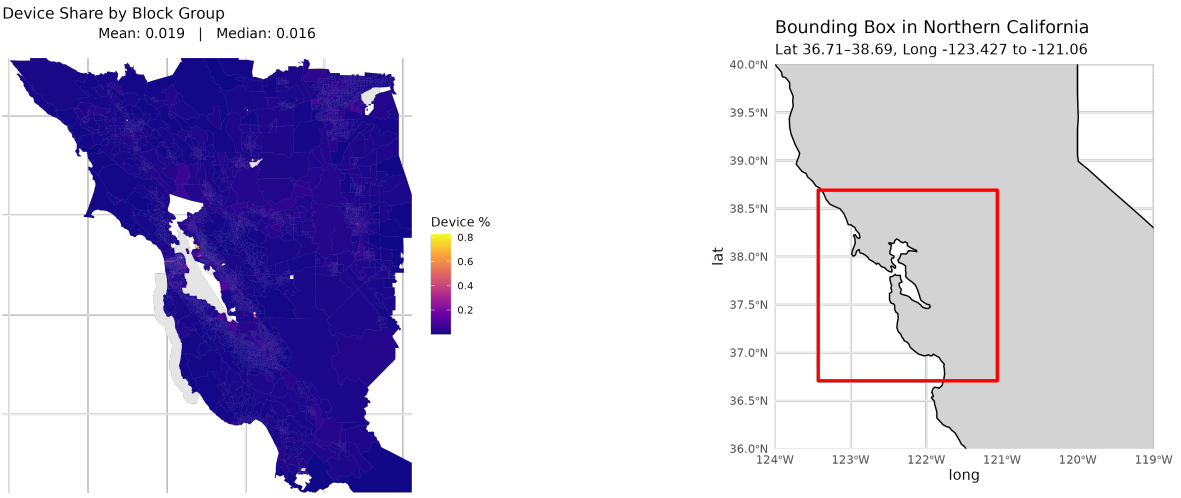
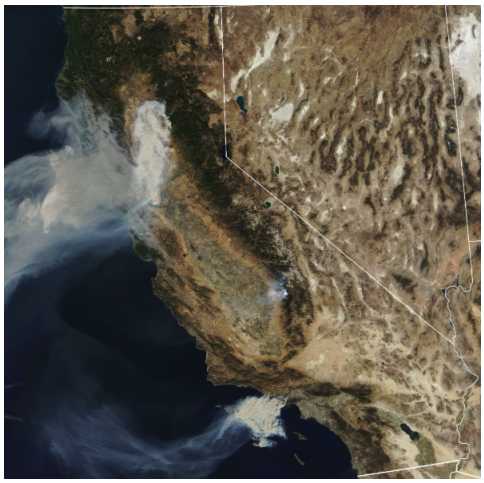


Table S1. Device count, workplace identification, and blockgroup socioeconomic characteristics

Panel A: Workplace Identification		
Identified workplace		
log(household median income)	0.080***	
	(0.002)	
log(# college educated)		0.047***
		(0.001)
Observations	175,956	182,759
Panel B: # of Devices by Blockgroup		
log(# identified devices in block group)		
log(household median income)	-0.512***	
	(0.014)	
log(# college educated)		-0.086***
		(0.009)
Observations	6737	7019

SE in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. In panel A "identified workplace" is an indicator of whether a workplace is identified for this device. In Panel B, unit is the block group.

Fig. S2. Satellite Image of California on November 9, 2018



Note: Image from NASA Worldview (1)

Table S2. Comparison to the American Time Use Survey Metrics

Panel A: American Time Use Survey Metrics

	mean	sd	min/max
hours inside	23.288	1.450	7/24
hours outside	0.712	1.450	0/17
hours home	18.057	4.692	0/24
hours worked (cond. on working)	6.916	3.285	0/23
Observations	4,162		

Panel B: Smartphone Data Time-Use Metrics

hours inside	18.177	5.221	0/24
hours outside	3.167	3.951	0/24
hours at home	14.798	6.717	0/24
hours at work (cond. on working)	7.111	3.988	0/24
daily total hours observed	21.344	2.717	15/24

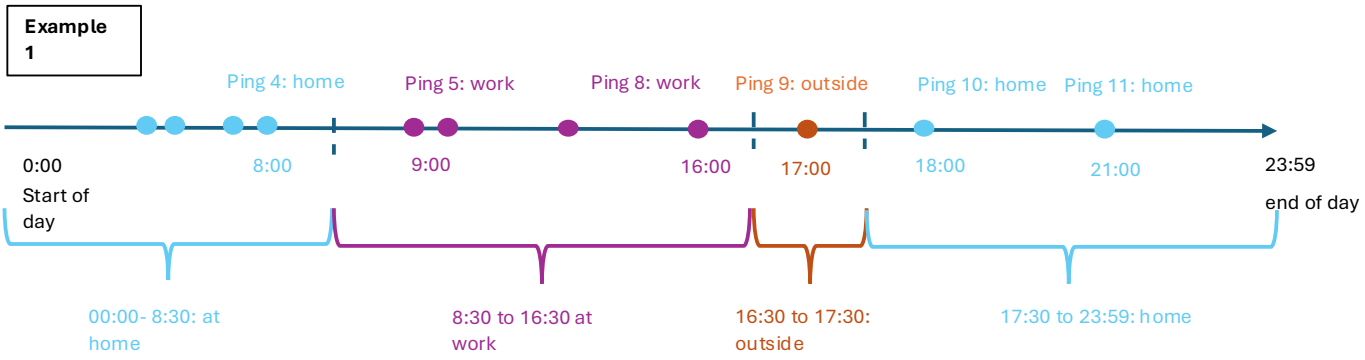
Units are hours. In ATUS, the number of hours observed in a day is mechanically 24 hours.

Table S3. Robustness - Outcome in Hours & Clustering at day × tract level

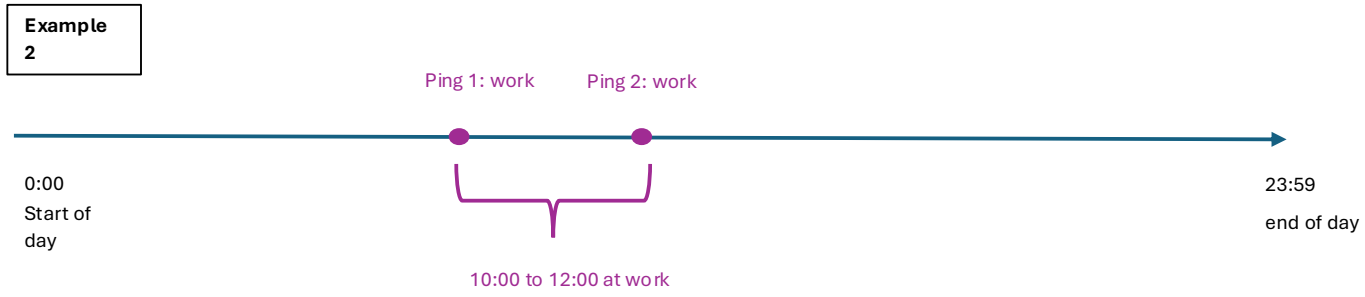
	Outcome in Hours			**Clustering at tract × day level**			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	hours at home	hours inside	hours at work	% time at home	% time inside	% time at work	log(Distance)
5-25 $\mu g/m^3$	0.066***	0.003	-0.125***	0.221***	-0.058	-0.567***	-0.029***
	(0.014)	(0.011)	(0.020)	(0.053)	(0.039)	(0.104)	(0.004)
25-50 $\mu g/m^3$	0.027	0.022	-0.128***	0.055	0.070	-0.550***	-0.006
	(0.026)	(0.022)	(0.040)	(0.108)	(0.079)	(0.195)	(0.008)
50-100 $\mu g/m^3$	0.248***	0.209***	-0.200***	0.889***	0.713***	-0.940***	-0.052***
	(0.021)	(0.017)	(0.030)	(0.083)	(0.059)	(0.143)	(0.006)
100-150 $\mu g/m^3$	0.236***	0.217***	-0.129**	0.934***	0.837***	-0.617**	-0.041***
	(0.035)	(0.028)	(0.059)	(0.139)	(0.097)	(0.296)	(0.011)
150+ $\mu g/m^3$	0.671***	0.431***	-0.320***	2.378***	1.123***	-1.882***	-0.022
	(0.054)	(0.042)	(0.090)	(0.265)	(0.160)	(0.428)	(0.017)
temp. and prec. controls	yes	yes	yes	yes	yes	yes	yes
Month FE, device FE	yes	yes	yes	yes	yes	yes	yes
day-of-week FE	yes	yes	yes	yes	yes	yes	yes
clustering level	device	device	device	tract × day	tract × day	tract × day	tract × day
# of devices	162,558	162,558	23,677	162,558	162,558	23,677	162,558
N	4,771,539	4,771,539	876,078	4,771,539	4,771,539	876,078	4,771,539
R^2	0.448	0.378	0.449	0.441	0.365	0.446	0.517

Notes: SE in parentheses and clustered at the device level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Only includes device-day observations that travel less than 300km during the day, whose home is not within 50km of an active wildfire, and for which we observe at least 10 pings, 15 hours of the day, and 10 hours with at least one ping. SEs are clustered at the day × tract level. The baseline category is days when smoke is between 0 and 5 μg/m^3.

Fig. S3. Examples of Methodology



Results: 24 hours divided between **15 hours at home, 8 hours at work, 1 hour outside**
➔ Observing more than 20 hours and more than 10 pings: makes it into the sample ✓



Results: 2 hours; **2 hours at work**
➔ Observing less than 20 hours and less than 10 pings: doesn't make it into the sample ✗

Table S4. Robustness - Alternative $PM_{2.5}$ bins

	(1) % time at home	(2) % time inside	(3) % time at work	(4) log(Distance)
25-50 $\mu g/m^3$	0.020	0.079	-0.472**	-0.001
	(0.108)	(0.082)	(0.191)	(0.008)
50-75 $\mu g/m^3$	0.757***	0.636***	-0.980***	-0.045***
	(0.097)	(0.075)	(0.152)	(0.007)
75-100 $\mu g/m^3$	1.131***	0.899***	-0.535**	-0.059***
	(0.135)	(0.098)	(0.273)	(0.010)
100-125 $\mu g/m^3$	1.062***	0.930***	-0.592	-0.072***
	(0.181)	(0.133)	(0.367)	(0.014)
125-150 $\mu g/m^3$	0.766***	0.750***	-0.509	-0.003
	(0.194)	(0.143)	(0.424)	(0.015)
150+ $\mu g/m^3$	2.367***	1.133***	-1.828***	-0.019
	(0.221)	(0.157)	(0.425)	(0.017)
temp. and prec. controls	yes	yes	yes	yes
Month FE, device FE, day-of-week FE	yes	yes	yes	yes
# of devices	162,558	162,558	23,677	162,558
N	4,771,539	4,771,539	876,078	4,771,539
R^2	0.441	0.365	0.446	0.517

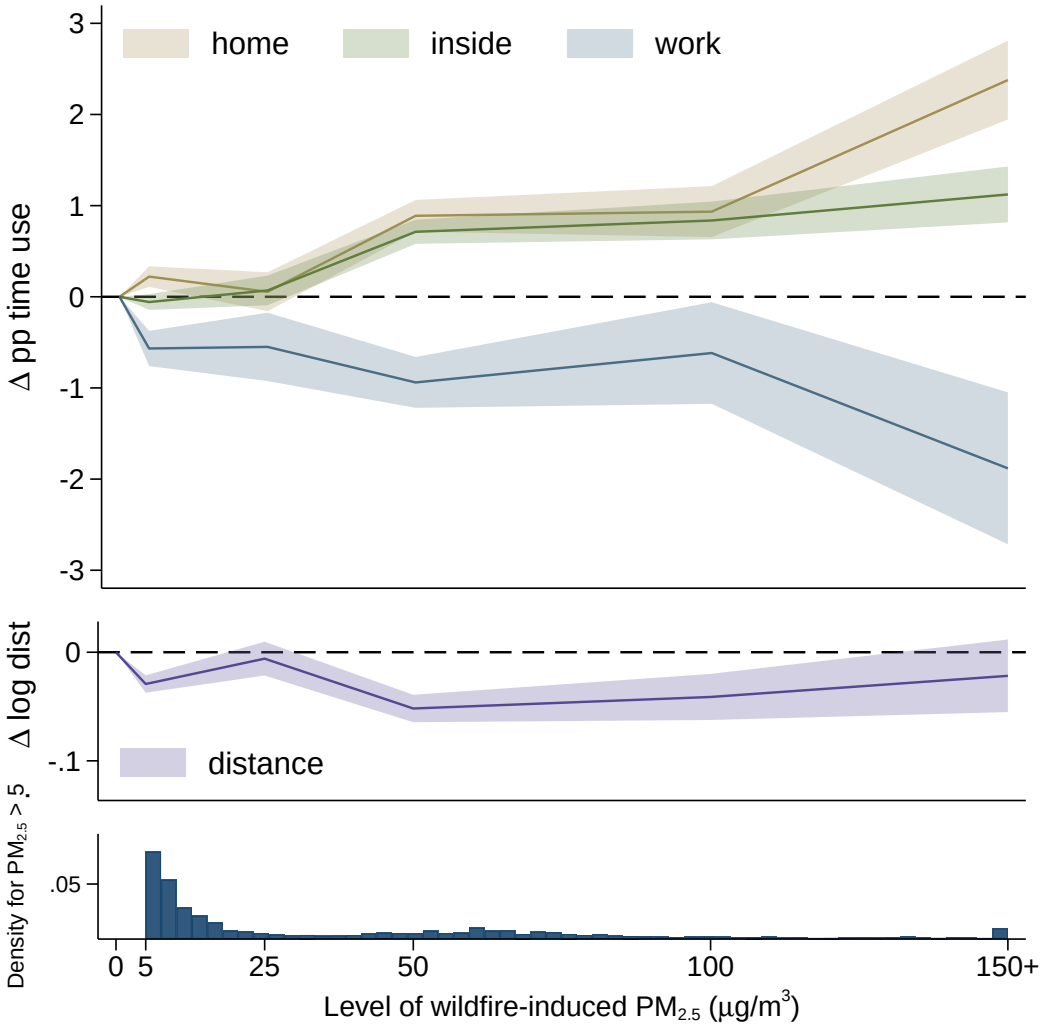
Notes: SE in parentheses and clustered at the device level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Only includes device-day observations that travel less than 300km during the day, whose home is not within 50km of an active wildfire, and for which we observe at least 10 pings, 15 hours of the day, and 10 hours with at least one ping.

Table S5. Temporary out-migration due to wildfire smoke

	(1)	(2)	(3)
weekly max dist. from home > ...	50km	100km	200km
smoke > 150 $\mu g/m^3$	0.022***	0.003**	-0.002
	(0.002)	(0.001)	(0.001)
smoke > 150 $\mu g/m^3$ × educ > 75th pct	-0.002	0.009*	0.009**
	(0.005)	(0.004)	(0.004)
smoke > 150 $\mu g/m^3$ × inc > 75th pct	0.001	0.004	0.011***
	(0.005)	(0.005)	(0.004)
weekly data	yes	yes	yes
month FE	yes	yes	yes
device FE	yes	yes	yes
temp. and prec. controls	yes	yes	yes
devices	260,808	260,808	260,808
N	7,660,529	7,660,529	7,660,529

SE in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. SE clustered at the device level. The dependent variable is an indicator capturing whether the maximum of the daily maximum distance from home (in km) for each device is above 50, 100, or 150 km. Estimating equation is presented in Equation 3 and more about methodology is presented in the Methods sections.

Fig. S4. Time Use Dose Responses to Wildfire Smoke — Binned Responses



Legend: The graph shows estimated coefficients for a binned specification with $PM_{2.5}$ bins of 5-25, 25-50, 50-100, 100-150, and above 150 $\mu g/m^3$ (coefficients reported in Table 2 in main text). Controls include quadratics for temperature and precipitation, day-of-week, device, and month fixed effects. Standard Errors are estimated for each coefficient and clustered at the device level. The density plot omits observations with smoke between 0 and 5 $\mu g/m^3$.

Table S6. Smoke, Rain, Income and Education

	Smoke			Precipitation	
	(1)	(2)	(3)	(4)	(5)
% time inside	smoke × inc.	smoke × educ.	3-way int.	prec × inc.	prec × educ.
smoke 50-150 $\mu g/m^3$	0.748***	0.704***	0.709***		
	(0.0710)	(0.0703)	(0.0743)		
smoke > 150 $\mu g/m^3$	1.019***	0.850***	0.839***		
	(0.170)	(0.176)	(0.182)		
smoke 50-150 $\mu g/m^3$ × income > 75th pct	-0.156		-0.0449		
	(0.151)		(0.219)		
smoke > 150 $\mu g/m^3$ × income > 75th pct	0.598		0.195		
	(0.552)		(0.710)		
smoke 50-150 $\mu g/m^3$ × education > 75th pct		0.0279	0.301		
		(0.151)	(0.224)		
smoke > 150 $\mu g/m^3$ × education > 75th pct		1.503***	1.574***		
		(0.448)	(0.512)		
smoke 50-150 $\mu g/m^3$ × education > 75th pct × income > 75th pct			-0.413		
			(0.352)		
smoke > 150 $\mu g/m^3$ × education > 75th pct × income > 75th pct			-0.344		
			(1.175)		
prec 10-30mm				0.506***	0.565***
				(0.0414)	(0.0408)
prec > 30mm				1.069***	0.963***
				(0.104)	(0.106)
prec 10-30mm × income > 75th pct				0.152	
				(0.0943)	
prec > 30mm × income > 75th pct				0.135	
				(0.238)	
prec 10-30mm × education > 75th pct					-0.0924
					(0.0992)
prec > 30mm × education > 75th pct					0.589***
					(0.228)
temp. and prec. controls	yes	yes	yes	no	no
smoke and prec. controls	no	no	no	yes	yes
Month FE, device FE, day-of-week FE	yes	yes	yes	yes	yes
# of devices	155,719	155,719	155,719	155,719	155,719
N	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730
R^2	0.364	0.364	0.364	0.364	0.364

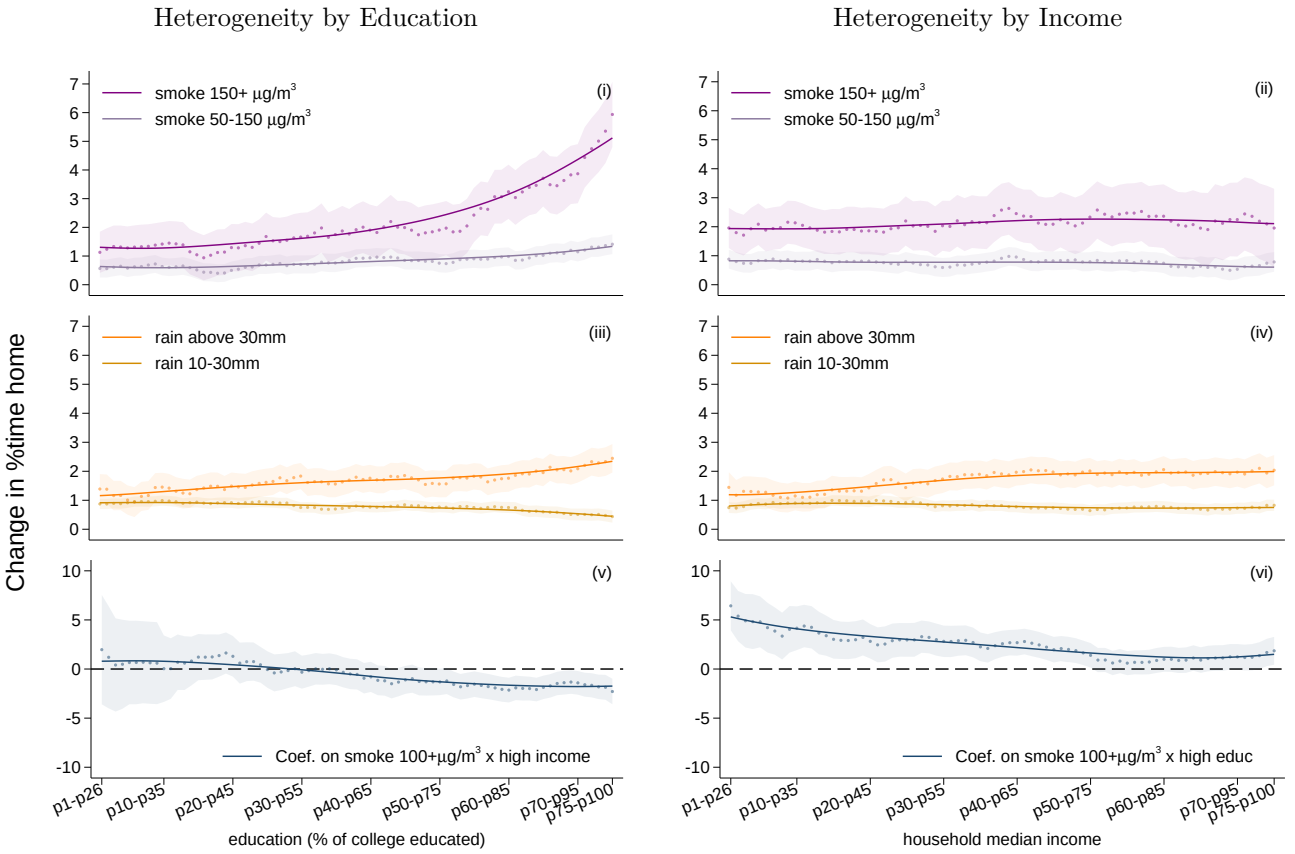
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. SE in parentheses and clustered at the device level. "Inc. > 75th pct" refers to devices identified as living in blockgroups that are in the top quartile of the income distribution based on median household income obtained in the ACS. "Educ. > 75th pct" refers to devices identified as living in block groups that are in the top quartile of the education distribution, based on the percentage of residents with bachelor degrees or higher in the ACS. Only includes device-day observations that travel less than 300km during the day, whose home is not within 50km of an active wildfire, and for which we observe at least 10 pings, 15 hours of the day, and 10 hours with at least one ping. The baseline category is pollution between 0 and 50 μg/m^3 for smoke, and days with rainfall between 0 and 10mm for responses to precipitation.

Table S7. Sensitivity of Role of Education to Smoke Thresholds

	(1)	(2)	(3)
indpt. var: indicator for smoke $PM_{2.5}$ being above... ($\mu g/m^3$)	50	100	150
smoke $>$ threshold	0.719***	0.572***	0.577***
	(0.073)	(0.104)	(0.178)
smoke $>$ threshold $\times$ educ $>$ 75th pct.	0.392*	1.196***	1.547***
	(0.221)	(0.354)	(0.507)
smoke $>$ threshold $\times$ inc $>$ 75th pct.	-0.039	-0.522	0.205
	(0.218)	(0.437)	(0.707)
(smoke $>$ threshold) $\times$ (inc $>$ 75th pct.) $\times$ (educ $>$ 75th pct.)	-0.481	-0.031	-0.303
	(0.349)	(0.654)	(1.170)
temp. and prec. controls	yes	yes	yes
Month FE, device FE, day-of-week FE	yes	yes	yes
# of devices	155,719	155,719	155,719
N	4,569,730	4,569,730	4,569,730
R^2	0.364	0.364	0.364

SE in parentheses and clustered at the device level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. SE clustered at the device level. Only includes device-day observations that travel less than 300km during the day, whose home is not within 50km of an active wildfire, and for which we observe at least 10 pings, 15 hours of the day, and 10 hours with at least one ping. Baseline category depends on the smoke threshold.

Fig. S5. Responses Across Socioeconomic Groups for Time at Home



Legend: Each plot represents plots coefficients for 75 regressions. In panels (i)-(iv), each point represents a regression coefficient from estimating Equation 2 with 3 smoke or precipitation bins and restricted to a sample according to education or income level from the block-group level ACS. For income we use median household income and for education the percentage of college-educated individuals. We split the sample in 75 quantiles of the income and education distributions, starting with the bottom 25% and incrementally moving to the top 25%. Panels (v) and (vi) adopt the same quantile-based approach but estimate Equation 2 with $k = 1$. For each quartile of the education distribution, we measure the impact of high income (v), and the impact of high education for each quartile of the income distribution (vi). The dependent variable is time spent at home. We fit a non-parametric locally smoothed curve between estimated coefficients and quartile rank

Table S10. Heterogeneity Across Different Socioeconomic Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
dep var: % time inside	Educ.	Income	Urban	Black	White	Asian	Hispanic	age $>$ 75	age $<$ 18
smoke $>$ 150 $\mu g/m^3$	0.605***	0.753***	-1.910	1.009***	0.692***	0.763***	0.964***	0.854***	1.054***
	(0.172)	(0.166)	(1.554)	(0.226)	(0.184)	(0.170)	(0.181)	(0.180)	(0.186)
smoke $>$ 150 $\mu g/m^3 \times$ het. var. $>$ 75th pct.	1.390***	0.620	2.761*	-0.400	0.482	0.405	-0.733**	-0.194	-0.761**
	(0.449)	(0.550)	(1.563)	(0.319)	(0.365)	(0.480)	(0.372)	(0.378)	(0.354)
temp. and prec. controls	yes	yes	yes	yes	yes	yes	yes	yes	yes
Month FE, device FE, day-of-week FE	yes	yes	yes	yes	yes	yes	yes	yes	yes
# of devices	155,719	155,719	155,719	155,719	155,719	155,719	155,719	155,719	155,719
N	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730	4,569,730
R^2	0.364	0.364	0.364	0.364	0.364	0.364	0.364	0.364	0.364

SE in parentheses and clustered at the device level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. SE clustered at the device level. Smoke $>$ 150 μg/m^3 is an indicator for extreme smoke pollution days. "het. var. $>$ 75th pct." is an indicator equal to 1 if the device is within the top quartile of the specific heterogeneity dimension listed at the top of each column, where each characteristic is assigned based on the CBG of the device home location. Only includes device-day observations that travel less than 300km during the day, whose home is not within 50km of an active wildfire, and for which we observe at least 10 pings, 15 hours of the day, and 10 hours with at least one ping. Sample is restricted to devices for which there is median block group household income information in the ACS. Baseline category is smoke 0-150μg/m^3.